

Module: 13- Networking with Windows Server

1. Discuss the role of Windows Firewall in Windows Server and how to configure it.

Ans: Windows Firewall is a security feature in Windows Server that helps protect the server from unauthorized access and malicious network traffic. It filters incoming and outgoing network connections based on defined security rules.

➤ Role of Windows Firewall

- Protects the server from hackers and viruses.
- Blocks unauthorized users and applications.
- Allows only trusted network traffic.
- Monitors inbound and outbound connections.
- Improves overall server security.

➤ Types of Firewall Profiles

1. **Domain Profile** – Used when connected to a domain network.
2. **Private Profile** – Used for trusted private networks.
3. **Public Profile** – Used for public or untrusted networks.

➤ Steps to Configure Windows Firewall

Using GUI

1. Open Control Panel.
2. Click System and Security.
3. Open Windows Defender Firewall.
4. Select Advanced Settings.
5. Choose:
 - Inbound Rules
 - Outbound Rules
6. Click New Rule.

7. Select rule type:

- Program
- Port
- Predefined
- Custom

8. Allow or block the connection.

9. Apply profile settings and finish configuration.

Using PowerShell

```
New-NetFirewallRule -DisplayName "Allow HTTP" -Direction Inbound -  
Protocol TCP -LocalPort 80 -Action Allow
```

Windows Firewall is an essential security tool in Windows Server that controls network traffic and protects servers from unauthorized access.

2. What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

Ans: NAT (Network Address Translation) is a networking technique used to convert private IP addresses into public IP addresses. It allows multiple devices in a private network to access the internet using a single public IP address.

Advantages of NAT

- Saves public IP addresses.
- Provides basic network security.
- Allows internet sharing.
- Hides internal network structure.

Types of NAT

1. Static NAT
2. Dynamic NAT

3. PAT (Port Address Translation)

Steps to Configure NAT in Windows Server

Install Remote Access Role

1. Open Server Manager.
2. Click Add Roles and Features.
3. Install:
 - Remote Access
 - Routing

Configure NAT

1. Open Routing and Remote Access (RRAS).
2. Right-click server name.
3. Select Configure and Enable Routing and Remote Access.
4. Choose NAT and LAN Routing.
5. Select internet-connected interface.
6. Enable NAT on the interface.
7. Finish setup and start RRAS service.

NAT helps organizations connect private networks to the internet securely and efficiently.

3. Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

Ans: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network settings to devices in a network.

Functions of DHCP

- Assigns IP addresses automatically.
- Provides subnet mask, gateway, and DNS information.

- Reduces manual configuration errors.
- Simplifies network management.

DHCP Components

- DHCP Server
- DHCP Client
- Scope
- Lease

Steps to Configure DHCP in Windows Server 2016

Install DHCP Role

1. Open Server Manager.
2. Click Add Roles and Features.
3. Select DHCP Server role.
4. Complete installation.

Authorize DHCP Server

1. Open notification flag in Server Manager.
2. Click Complete DHCP Configuration.
3. Authorize server in Active Directory.

Create DHCP Scope

1. Open DHCP Console.
2. Expand IPv4.
3. Right-click and choose New Scope.
4. Enter:
 - Scope name
 - IP address range
 - Subnet mask
5. Configure:

- Default gateway
- DNS server
- Lease duration

6. Activate the scope.

DHCP automates IP address management and reduces administrative workload in Windows Server networks.

4. Describe the process of configuring DNS (Domain Name System) in Windows Server.

Ans: DNS (Domain Name System) converts domain names into IP addresses. It is one of the most important services in Windows Server.

Functions of DNS

- Name resolution
- Supports Active Directory
- Helps clients locate servers and websites

Steps to Configure DNS in Windows Server

Install DNS Role

1. Open Server Manager.
2. Click Add Roles and Features.
3. Select DNS Server role.
4. Install the role.

Configure DNS Zone

1. Open DNS Manager.
2. Expand server name.
3. Right-click Forward Lookup Zones.
4. Select New Zone.
5. Choose:
 - Primary Zone

6. Enter zone name (example: company.local).
7. Finish the wizard.

Add Host Records

1. Right-click zone.
2. Select New Host (A or AAAA).
3. Enter:
 - Host name
 - IP address
4. Click **Add Host**.

Types of DNS Zones

- Primary Zone
- Secondary Zone
- Stub Zone

DNS is essential for communication in Windows Server networks because it translates names into IP addresses.

5. What is Server Manager, and how do you use it to manage servers in Windows Server?

Ans: Server Manager is a management tool used to install, configure, and monitor server roles and features in Windows Server.

Functions of Server Manager

- Add or remove roles and features.
- Manage local and remote servers.
- Monitor server performance.
- Configure services.
- Access management tools.

Features of Server Manager

- Dashboard interface
- Multi-server management
- Role-based administration
- Performance monitoring

How to Use Server Manager

Open Server Manager

- Automatically opens after login.
- Or search from Start Menu.

Common Tasks

1. Add Roles and Features
2. Manage DNS, DHCP, AD DS
3. Configure local server settings
4. Check server events and alerts
5. Manage storage and services

Server Manager simplifies administration tasks and provides centralized management for Windows Servers.

6. Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it.

Ans: Remote Desktop Services (RDS) allows users to remotely access desktops, applications, and servers over a network.

Roles of RDS

- Remote access to servers
- Centralized application management
- Supports remote work
- Allows multiple users to access one server

Components of RDS

1. Remote Desktop Session Host (RDSH)
2. Remote Desktop Gateway
3. Remote Desktop Licensing
4. Remote Desktop Web Access

Steps to Configure RDS

Install RDS Role

1. Open Server Manager.
2. Click Add Roles and Features.
3. Select Remote Desktop Services Installation.
4. Choose:
 - Quick Start
 - Standard Deployment

Configure Session Host

1. Select server for RDS Session Host.
2. Install required services.
3. Restart server if required.

Enable Remote Desktop

1. Open System Properties.
2. Select Remote tab.
3. Enable:
 - Allow remote connections to this computer

Configure Licensing

1. Open RD Licensing Manager.
2. Activate license server.
3. Install RDS CAL licenses.

Advantages of RDS

- Easy remote access
- Reduced hardware cost
- Centralized management
- Better resource utilization

Remote Desktop Services allows secure remote access and centralized application management in Windows Server 2016/2019.